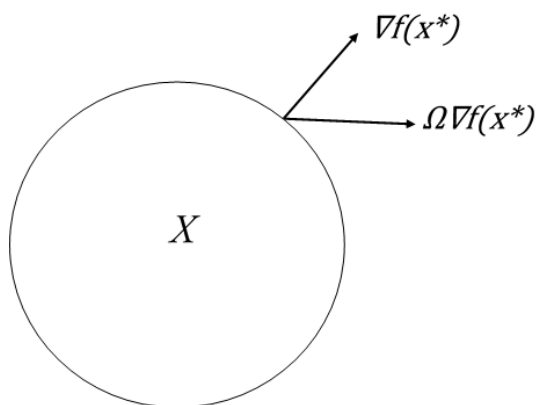


ECE 647: Lecture 27

Scaled projection



We may be tempted to do:

$$x((t+1)) = [x(t) - \gamma \Omega \nabla f(x(t))]^\dagger \quad (1)$$

where Ω is a positive-definite matrix.

However, in general the scaled version will not converge to the optimal point.

We will have to redefine $[x]^\dagger$ according to Ω , *i.e.*, ,

$$[x]^\dagger \Omega = \operatorname{argmin}(\delta - x)^T \Omega^{-1} (\delta - x) \quad (2)$$

In that case,

$$x((t+1)) = [x(t) - \gamma \Omega \nabla f(x(t))]^\dagger \Omega \quad (3)$$

will converge to x^*

Doesn't matter if X is a quadrant set & Ω is diagonal. (Bertsekas P214-215)

Subgradients

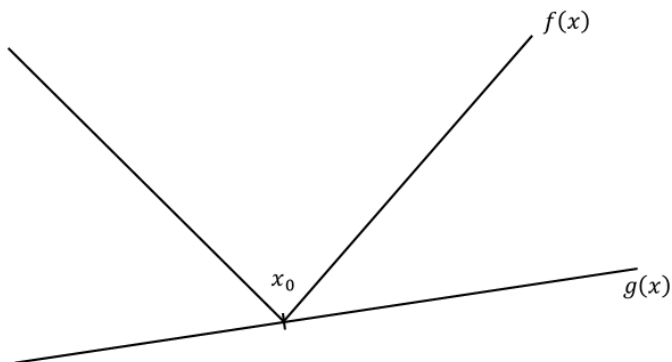
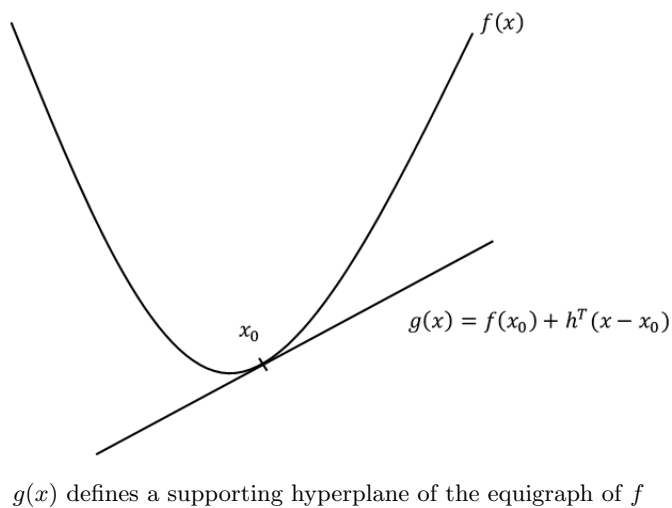
What if ∇f is not Lipschitz, or if the convex function f is not even differentiable?

We will need to use the notion of sub-gradients.

Define the subgradient of f at x_0 as a vector h such that

$$f(x) \geq f(x_0) + h^T(x - x_0) \quad (4)$$

for all $x \in \mathbb{C}$.



The set of all subgradients of f at x_0 is called the sub-differential of f at x_0 , denoted by $\partial f(x_0)$.

Fact 1. *A convex function f has non-empty sub-differentials at any*

$$x \in \text{int}(\text{dom } f). \quad (5)$$

(Will revisit later when we study separation theorem.)

If f is differentiable, then

$$\partial f(x_0) = \{f'(x_0)\} \quad (6)$$

First-order necessary and sufficient conditions for unconstrained problems:

$$0 \in \partial f(\bar{x}) \quad (7)$$

$$\Leftrightarrow f(x) \geq f(\bar{x}) + o(x - \bar{x}) \quad (8)$$

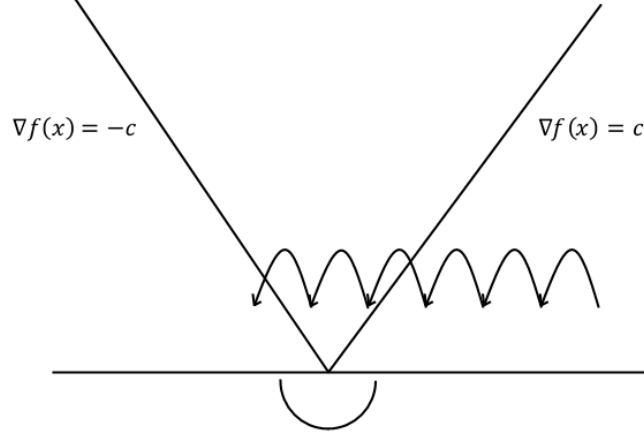
Subgradient Descent

Subgradient-descent algorithm:

$$x(t+1) = x(t) - \gamma \nabla f(x(t)) \quad (9)$$

where $\nabla f(x(t))$ is a subgradient.

However, it may not converge to a single point.



For any positive stepsize, $x(t)$ may oscillate around x^* .

As the stepsize goes to zero, the deviation will go to zero.

Let Ω be the set of points x such that

$$f(x) = f(x^*) \quad (10)$$

Let Ω_ε be the set of points such that

$$f(x) \leq f(x^*) + \varepsilon \quad (11)$$

Assume that $\|\nabla f(x(t))\|$ is bonded by M .

Recall that

$$\|x(t+1) - x^*\|^2 \leq \|x(t) - x^*\|^2 + \gamma^2 \|\nabla f(x(t))\|^2 - 2\gamma \nabla f(x(t))^T (x(t) - x^*) \quad (12)$$

Since f is convex,

$$f(x^*) \geq f(x(t)) + \nabla f(x(t))^T (x^* - x(t)) \quad (13)$$

Hence, if $x(t) \notin \Omega_\varepsilon$, then

$$\nabla f(x(t))^T (x^* - x(t)) \leq f(x^*) - f(x(t)) \leq -\varepsilon \quad (14)$$

Let $\gamma < \frac{\varepsilon}{M^2} \Rightarrow \gamma\varepsilon \geq \gamma^2 M^2$, then

$$\|x(t+1) - x^*\|^2 \leq \|x(t) - x^*\|^2 - \gamma\varepsilon \quad (15)$$

$\Rightarrow x(t)$ must enter Ω_ε eventually.

Once $x(t) \in \Omega_\varepsilon$, then if $\gamma < \frac{\varepsilon}{M}$,

$$\Rightarrow \|x(t+1) - x(t)\| \leq \|\gamma \nabla f(x)\| \leq \varepsilon \quad (16)$$

$\Rightarrow x(t+1)$ cannot be far from Ω_ε .

We have proven the following:

Theorem 1. *Assume that f is convex and its subgradients are bounded. Consider the subgradient descent algorithm:*

$$x(t+1) = x(t) - \gamma \nabla f(x(t)) \quad (17)$$

where $\nabla f(x(t))$ is a sub-gradient of f at $x(t)$. Then for any $\varepsilon > 0$, there exists a stepsize γ such that

$$d(x(t), \Omega) < \varepsilon, \forall t \geq T \quad (18)$$

A similar result can be stated for constrained problems.

Diminishing stepsize

Alternatively, if we can let the stepsize go to zero in an appropriate fashion, then we can still claim convergence.

Theorem 2. *Assume that f is convex & continuous, and its subgradient is bounded. Consider the subgradient descent algorithm:*

$$x(t+1) = x(t) - \gamma_t \nabla f(x(t)) \quad (19)$$

where $\nabla f(x(t))$ is a subgradient of f at $x(t)$.

Further, assume that

$$\sum_{t=1}^{+\infty} \gamma_t \rightarrow +\infty \quad (20)$$

$$\sum_{t=1}^{+\infty} \gamma_t^2 < +\infty \quad (21)$$

Then as $t \rightarrow +\infty$, $x(t)$ converges to x_0 , and $\nabla f(x_0) = 0$, provided that there exists at least one point x^* with $\nabla f(x^*) = 0$

Proof:

$$\|x(t+1) - x^*\|^2 \leq \|x(t) - x^*\|^2 + \gamma_t^2 \|\nabla f(x(t))\|^2 - 2\gamma_t (\nabla f(x(t)))^T (x(t) - x^*) \quad (22)$$

If $\nabla f(x(t))$ is bounded by M , then

$$\|x(t+1) - x^*\|^2 \leq \|x(0) - x^*\|^2 + \sum_{k=1}^t \gamma_k^2 M^2 - 2 \sum_{k=1}^t \gamma_k \nabla f(x(k))^T (x(k) - x^*) \quad (23)$$

Note that

$$\sum_{k=1}^{+\infty} \gamma_k^2 M^2 < +\infty \quad (24)$$

Hence:

$$\sum_{k=1}^{+\infty} \gamma_k \nabla f(x(k))^T (x(k) - x^*) < +\infty \quad (25)$$

Further, note that

$$\nabla f(x(t))^T (x(t) - x^*) \geq f(x(t)) - f(x^*) \geq 0 \quad (26)$$

Hence, using $\sum_{k=1}^{\infty} \gamma_k = +\infty$, there must exist a sub-sequence t_n such that

$$\lim_{n \rightarrow +\infty} \nabla f(x(t_n))(x(t_n) - x^*) = 0 \quad (27)$$

We can verify that $x(t_n)$ is a bounded sequence.

Hence, there exists a limit point x_0 .

First we want to show that $f(x_0)$ is optimal.

Note that

$$f(x^*) \geq f(x(t_n)) + \nabla f(x(t_n))(x^* - x(t_n)) \quad (28)$$

$$\Rightarrow \nabla f(x(t_n))(x(t_n) - x^*) \geq f(x(t_n)) - f(x^*) \quad (29)$$

Hence,

$$0 \leq \lim_{n \rightarrow +\infty} f(x(t_n)) - f(x^*) \leq \lim_{n \rightarrow +\infty} \nabla f(x(t_n))(x(t_n) - x^*) = 0 \quad (30)$$

$$f(x_0) = \lim_{n \rightarrow +\infty} f(x(t_n)) = f(x^*) \quad (31)$$

Second, we want to show that $x(t) \rightarrow x_0$.

Replacing x^* by x_0 , we have

$$\|x(t+1) - x_0\|^2 \leq \|x(t) - x_0\|^2 + \gamma_t^2 M^2 \quad (32)$$

Summing over $t > t_n$

$$\|x(t+1) - x_0\|^2 \leq \|x(t_n) - x_0\|^2 + \sum_{k=t_n+1}^t \gamma_k^2 M^2 \quad (33)$$

$$(\sum_{k=t_n+1}^t \gamma_k^2 M^2 \rightarrow 0 \text{ as } t_n \rightarrow +\infty)$$

So,

$$\lim \|x(t) - x_0\|^2 \leq \lim \|x(t_n) - x_0\|^2 = 0 \quad (34)$$